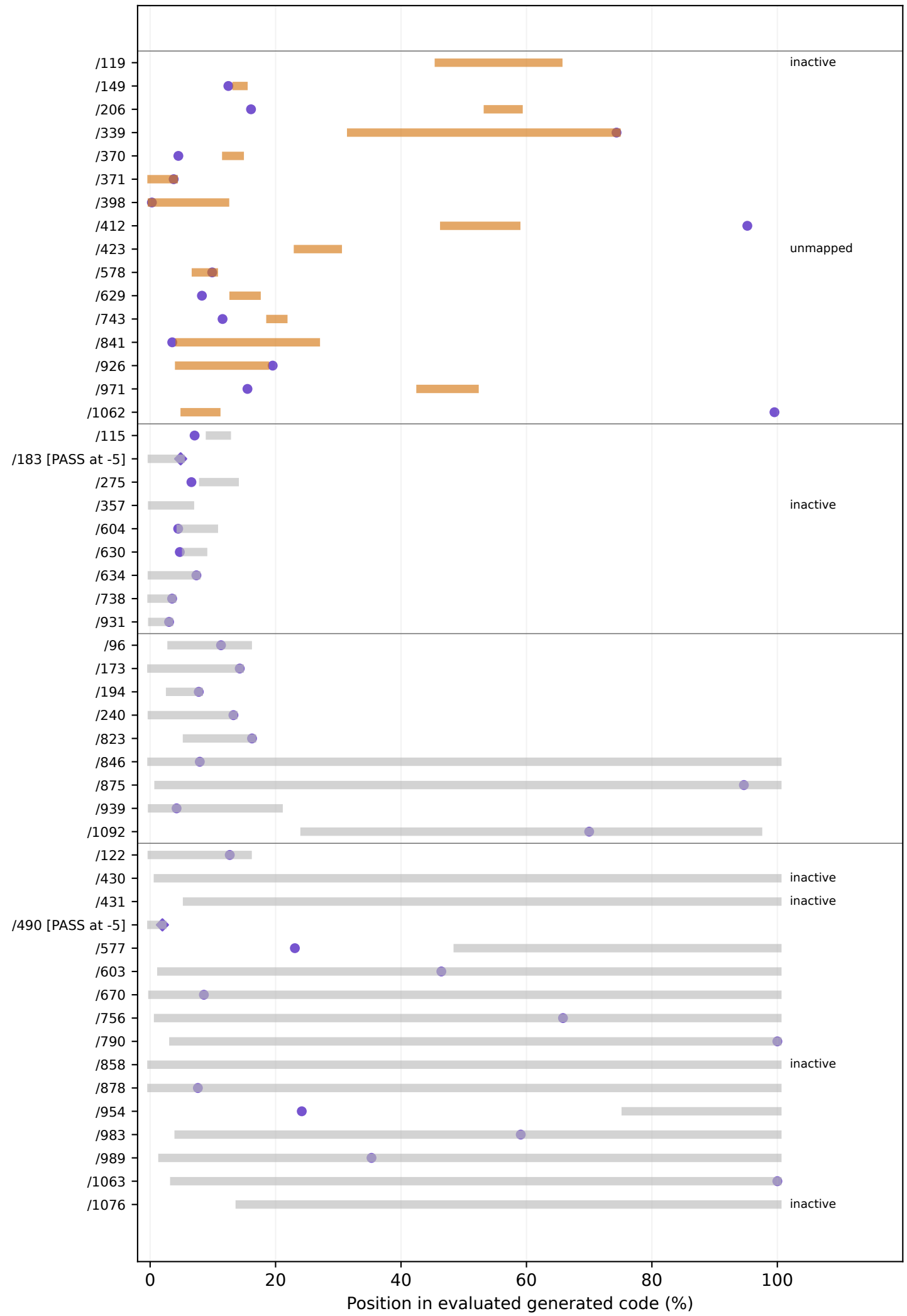
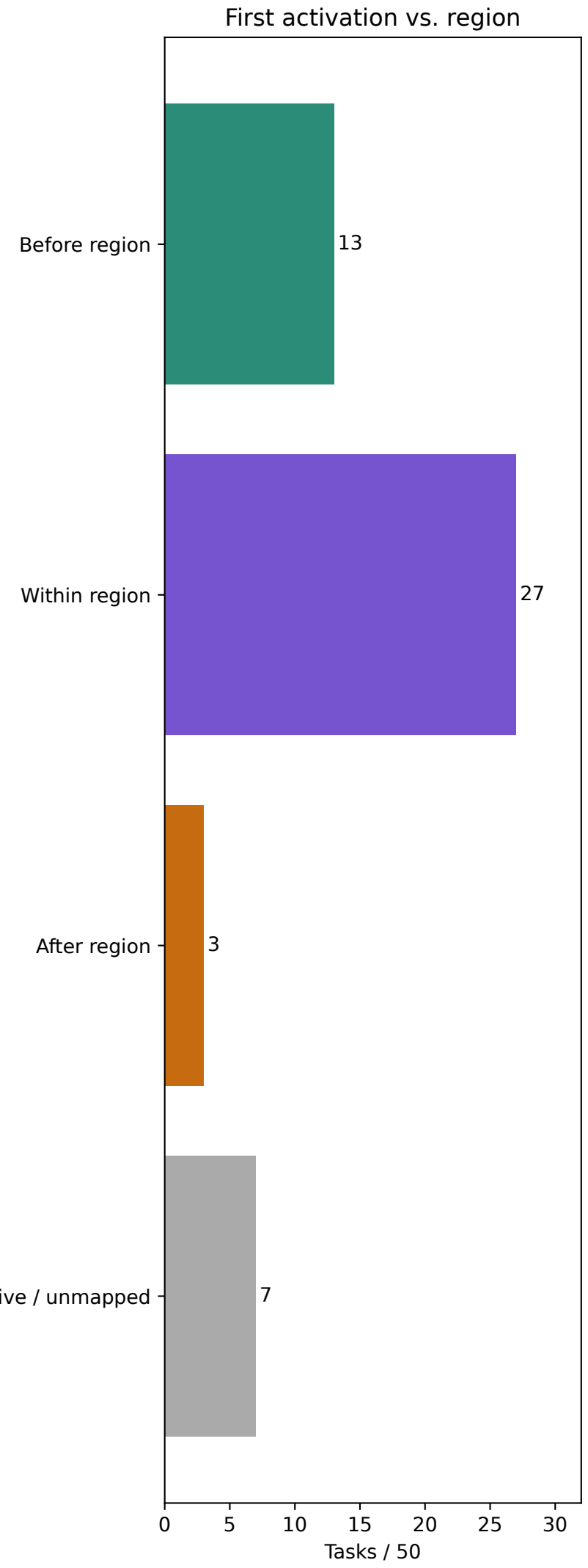
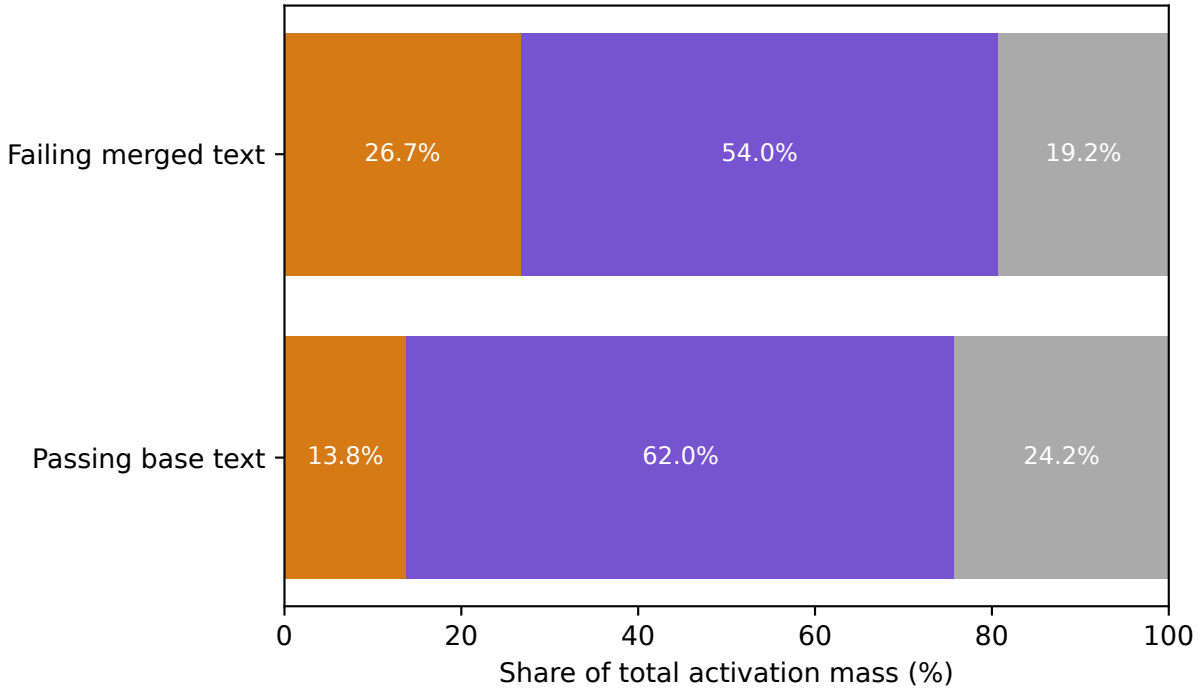
4309 timing in 50 failing merged outputs

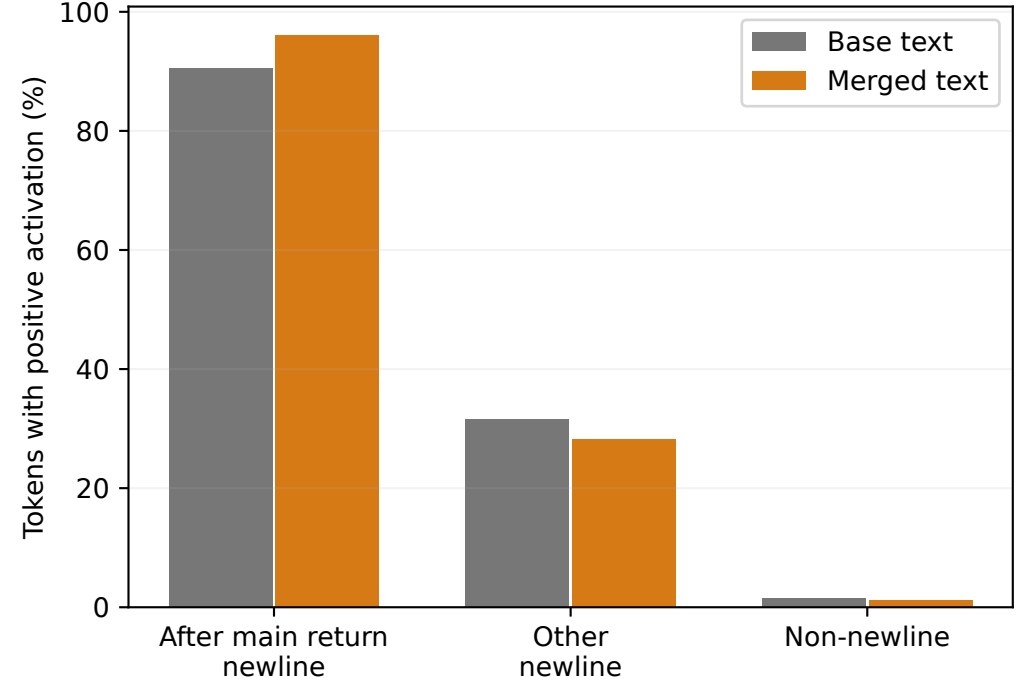




Purple: first natural activation · region: gray (proposed), orange (runtime site) · diamonds: two corrected tasks
Rows grouped: runtime sites / proposed lines / proposed intervals / omission regions. These are NOT validated error onsets.

Does 4309 prefer return-line boundaries?





50 task pairs · same shared feature on different generated texts · pooled descriptive statistics, not independent token-level tests.

Natural first-activation examples: return boundaries and counterexamples

BigCodeBench/183 | PASS at alpha -5

```
22      True
23      """
24      return HttpResponse(json.dumps(data), status=200)* [newline]
```

First activation: newline_after_main_return | z=4.552
Timing vs. region: same_token_or_within_region

BigCodeBench/490 | PASS at alpha -5

```
26      'Emma'
27      """
28      return xmltodict.parse(s)* [newline]
```

First activation: newline_after_main_return | z=3.357
Timing vs. region: same_token_or_within_region

BigCodeBench/634 | FAIL at alpha -5

```
32      ModeResult(mode=array([1]), count=array([5]))
33      """
34      return stats.mode(itertools.chain(input_list, input_list, input_list,
input_list))* [newline]
```

First activation: newline_after_main_return | z=4.055
Timing vs. region: same_token_or_within_region

BigCodeBench/115 | FAIL at alpha -5

```
30      """
31      if len(numbers) == 0:
32          raise ValueError('task_func() requires a non-empty list of
numbers')* [newline]
33      else:
```

First activation: other_newline | z=13.332
Timing vs. region: before

BigCodeBench/629 | FAIL at alpha -5

```
28    # Create output directory if it does not exist.
29     if not os.path.exists(output_dir):
30         os.mkdir(output_dir)* [newline]
31
```

First activation: other_newline | z=7.036
Timing vs. region: before

BigCodeBench/122 | FAIL at alpha -5

```
29      my_list = np.array(my_list)
30      random_number = random.randint(0, 100)
31      my_list = np.append(my_list, random_number)* [newline]
32      return np.array(my_list)
```

First activation: other_newline | z=1.814
Timing vs. region: same_token_or_within_region

BigCodeBench/206 | FAIL at alpha -5

```
36      """
37      if not os.path.exists(file_name):
38          raise FileNotFoundError(file_name*)
39      with open(file_name, 'r', newline='') as csvfile:
```

First activation: non_newline | z=9.161
Timing vs. region: before

BigCodeBench/96 | FAIL at alpha -5

```
36      with open(csv_file, "r") as f:
37          data = f.read().split(csv_delimiter)
38          return [ (word, wordcount) for wordcount, word in
Counter(data).items() ]* [newline]
39          sorted(operator.itemgetter(1), operator.itemgetter(0))
```

First activation: newline_after_main_return | z=2.235
Timing vs. region: same_token_or_within_region

Exact frozen merged-text snippets; * marks the token start (newline shown explicitly).
Includes both corrected cases, failed expansion, before-region cases, other-newline and non-newline counterexamples.
Additional examples selected by ascending task ID within the stated category; not a representative random sample.